GP-RareLink

Reading the rare-disease literature with a language model, and checking its work

Interim overview for collaborators · 2026-11-09

Rare Genomics Institute with the Stanford Open Virtual Assistant Lab · funded by Anthropic's AI for Science program

A variant published once, and then lost

When a child with a rare disease is finally diagnosed, the responsible variant is often reported in a single case report and never submitted to a public database. The laboratory testing the next child searches ClinVar and finds nothing.

The knowledge exists. It is scattered across thousands of papers, in prose and tables no database can query, and hand curation has never kept up.

Alexander disease is the example on the right: a 2024 review counts the published variants and patients; ClinVar holds a fraction of them.

182
distinct causative variants published for Alexander disease (2024 review)

550
published patients with Alexander disease

76
pathogenic or likely-pathogenic ClinVar records for the gene (release 2026-10-12)

The question, and the rules behind it

Can a language model, reading with strict rules, do the first pass of curation: which variant, in which person, with which clinical features, each value tied to the sentence it came from, at an accuracy we can measure rather than assume?

1 Quote everything

A value without a verbatim quote is not allowed. A script looks up every quote in the paper. "Not reported" is a valid answer; a paper's silence never becomes a "no".

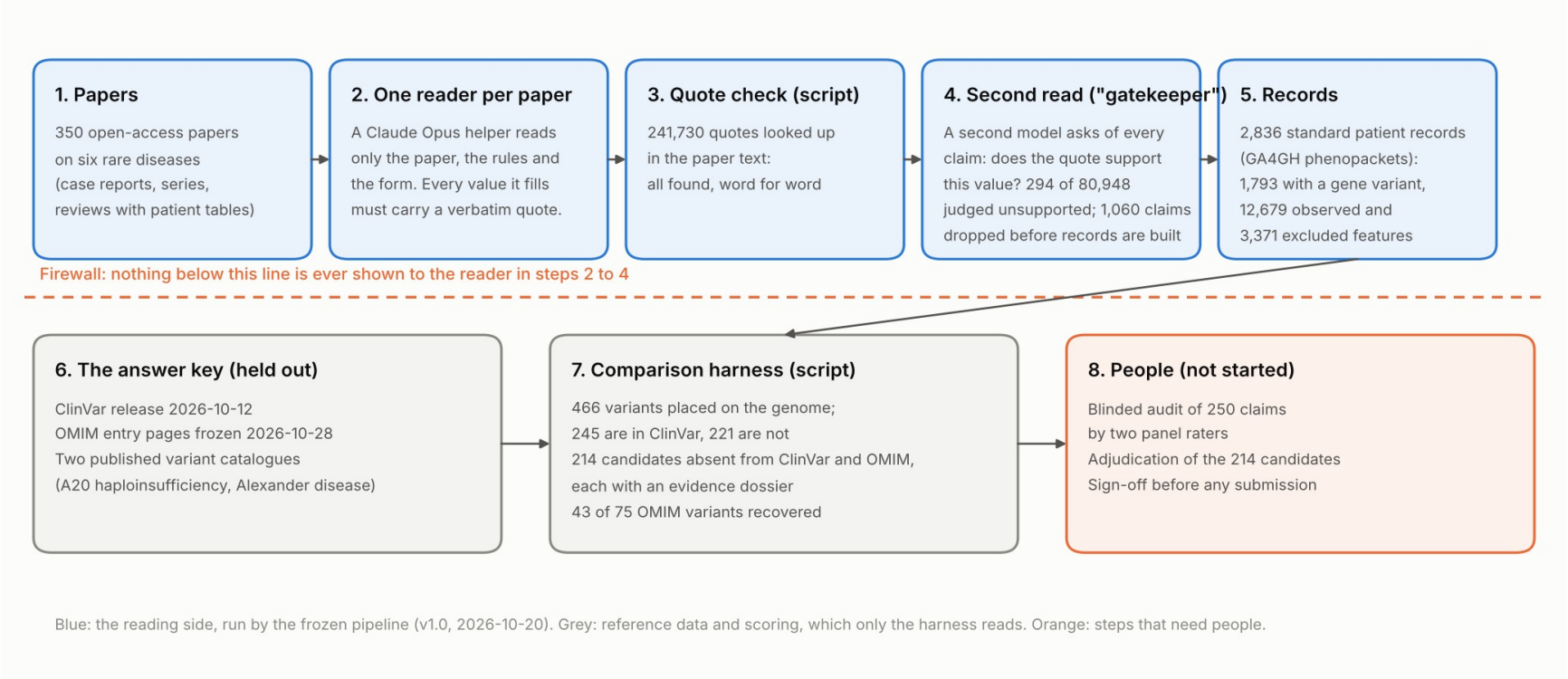
2 Hold out the answer key

ClinVar and OMIM are never shown to the model. Code that reads papers cannot read them; a test fails if it could. They are compared afterwards, by a separate harness.

3 Decide the rules first

Prompts, forms and the analysis plan were frozen on 2026-10-20, after a one-disease pilot. Changes since are logged as numbered deviations.

How it works: read, quote, check, compare



Blue: the reading side, run by the frozen pipeline (v1.0, 2026-10-20). Grey: reference data and scoring, which only the harness reads. Orange: steps that need people.

Six diseases, six organ systems

Disease	Gene	Inheritance	Papers read (open share)	Individuals	With a variant	Variants placed	Candidates	OMIM variants recovered
HA20*	*TNFAIP3*	AD	85 (53%)	769	558	103	64	8 of 9
SSADH	*ALDH5A1*	AR	40 (35%)	363	243	78	10	4 of 7
Tangier	*ABCA1*	AR	31 (14%)	235	65	16	7	2 of 19
LAD-I	*ITGB2*	AR	42 (22%)	508	328	129	82	12 of 14
AxD	*GFAP*	AD / de novo	69 (23%)	456	318	67	17	9 of 13
Danon	*LAMP2*	X-linked dominant	83 (39%)	505	281	73	34	8 of 13

*A20 haploinsufficiency (HA20) developed the pipeline; the other five are the confirmatory set. Only open-access papers were read. "Placed" = written unambiguously on the GRCh38 genome, which is what makes a variant comparable with ClinVar. "Candidates" = absent from ClinVar and OMIM, awaiting adjudication.

The numbers so far

350

open-access papers read, six diseases

2,836

individual records (GA4GH phenopackets), every value with a quote

241,730

quotes checked against the paper text: all found, word for word

466

distinct variants placed on the genome

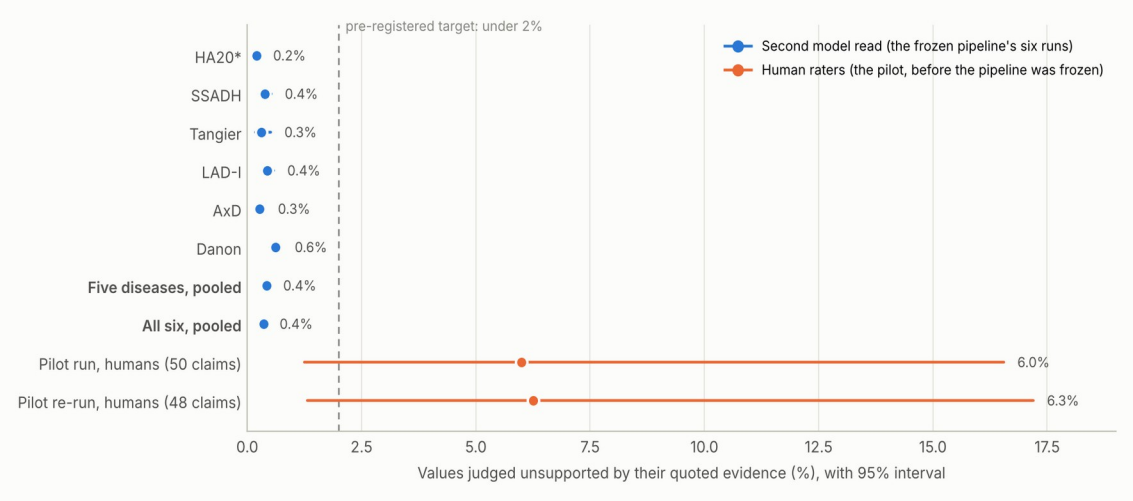
214

variants in neither ClinVar nor OMIM, each with an evidence dossier

43 of 75

variants OMIM lists for these diseases recovered (57%)

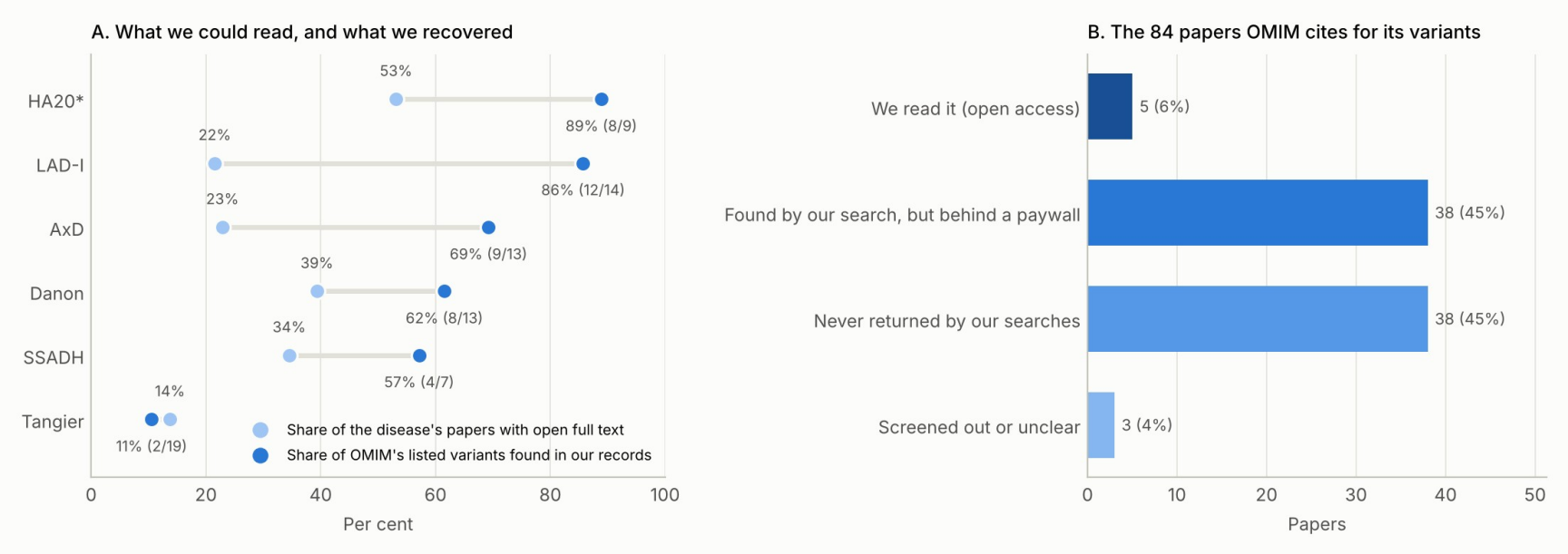
Finding 1: receipts hold, decisive check pending



Blue: the gatekeeper's verdicts per disease run. Orange: two human audits of the pilot's output. Dashed line: the 2% target.

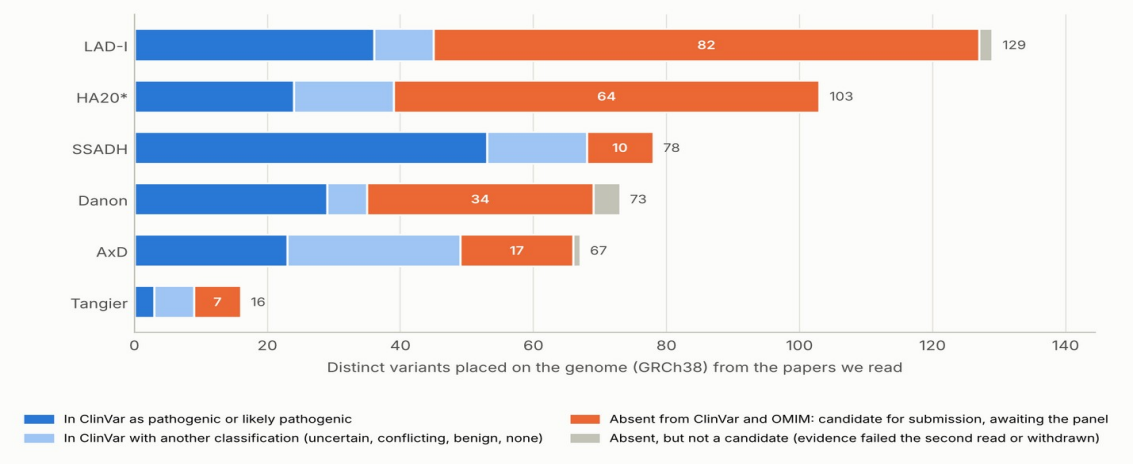
- Second model read: 0.43% of values unsupported on the five confirmatory diseases (0.36% with HA20).
- Human raters in the pilot, before the fixes were frozen: 6.0% and 6.3%.
- The pre-registered test is a blinded audit of 250 claims by two panel members. It is built; no claim has been rated.
- **Until then, the machine's rate is a filter's reading, not an accuracy.**

Finding 2: we find what we can read



Left: the share of each disease's papers with open full text (light) beside the share of OMIM's listed variants we recovered (dark). Right: of the 84 papers OMIM cites for those variants, we could read 5; 38 are behind paywalls and 38 were never returned by our searches, mostly pre-1990 papers under earlier names for the disease. Where the source paper was read, the variant was in our records 8 times out of 14.

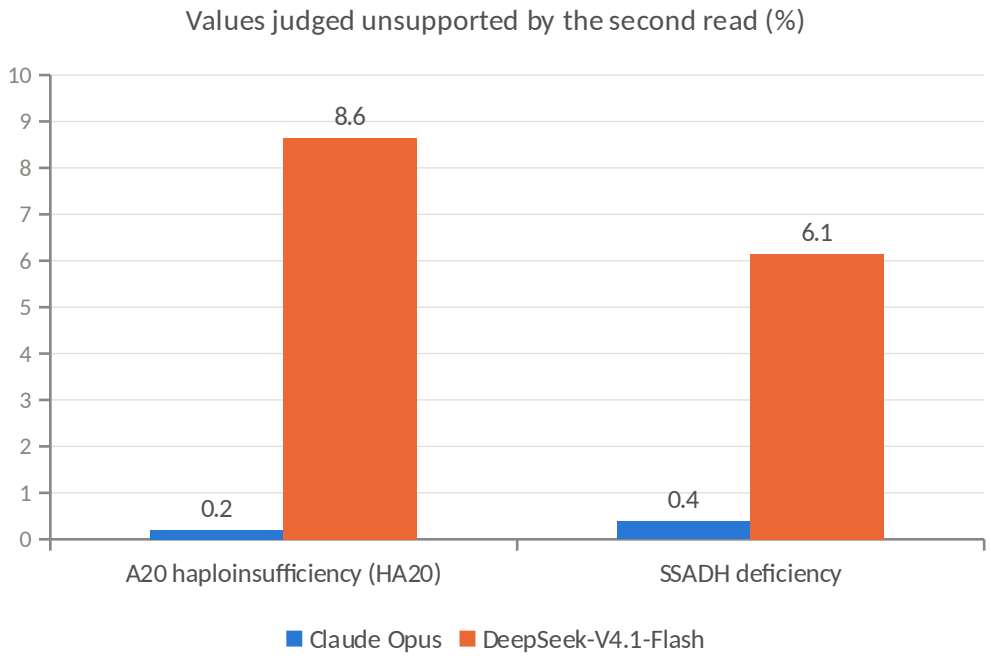
Finding 3: a real gap in the databases



Each bar: one disease's genome-placed variants. Orange: candidates awaiting the panel.

- 466 variants placed on the genome; 245 are in ClinVar, 168 of them pathogenic or likely pathogenic.
- 221 are absent from ClinVar. After dropping unsupported ones, checking OMIM and withdrawing one normalization error of our own: 214 candidates.
- 53 more sit in ClinVar as uncertain or conflicting while papers call them pathogenic; 15 with strong evidence.
- Every candidate has a dossier: papers, quotes, individuals, the authors' classification, and flags for the panel. Nothing is submitted without adjudication.

Finding 4: same variants, lost receipts, lower cost



- Same papers, prompts, forms and scoring. The open model found the same variants 89% to 95% of the time (shared over union).
- About 1/15 of the cost per paper ($0.07 vs $1.08 median) and 7 times the wall-clock.
- 3.0% to 3.6% of its values came with no quote at all, which the rules forbid.
- The judge is itself Claude, so the comparison is not neutral until the planned blinded human audit of both outputs.

Two more checks: repeats and memorization

A Run-to-run stability

A tenth of the papers (38) were run a second time.

Same checklist answers 97% to 100% of the time.

Same variants 83% to 100% of the time.

Where the runs differed, it was mostly over how many individuals a cohort paper describes.

B Memorization probe

180 papers were probed, stratified around the model's training cutoff: asked about each paper without its text, and asked to continue a passage.

In the 10 papers published after the cutoff, neither Claude nor the open model produced a specific recall or a correct variant.

The probe is small and will grow, but it gives no sign of memorization.

What it means

1 Accuracy

Records with complete provenance, stable across repeats, and a second read that removed 1,060 claims before records were built. What is missing is a human-rated error rate for the frozen pipeline. The 250-claim audit provides it and costs no model time.

2 Coverage

Reading is not the bottleneck; access is. With 14% to 53% of each corpus open, completeness and the candidate list are lower bounds. A decision on institutional or text-mining access would change these results more than any change to the model.

3 The answer key

Databases curated by people deserve the same scrutiny. OMIM lists selected examples, not a catalogue; two of its entries link to ClinVar records that contradict their text. The OMIM check caught one of our own normalization errors, now fixed in the pipeline.

What we need from you

Role	What you would do	Effort	What it unlocks
Audit rater	Rate 250 extracted values against their quoted evidence on a blinded web page; two raters and a tie-breaker	10 to 15 hours per rater	The primary outcome: the first true accuracy figure
Adjudicator	Review the 214 candidate variants from their dossiers and decide whether the evidence supports a submission	10 to 15 minutes per candidate	ClinVar literature-only submissions and OMIM candidates
Cohort reader	Fill a comparator sheet of clinical-feature frequencies from a natural-history cohort paper, without looking at our records	1 to 2 hours per sheet	The ascertainment-bias arm
Methods or domain reviewer	Read the short paper or the full report and tell us what is wrong, missing or unclear, from your field's point of view	1 to 3 hours	A better study before anything is submitted

Tell us who you are and how you would like to contribute through the form linked on the dashboard. Everything is published literature; there is no patient data.

Thank you

The dashboard holds the short paper, the full report and supplement, this deck, the overview video, the candidate list and the form.

https://cjimmylin.github.io/gp-rarelink/

Every number in this deck comes from the project repository at commit 8d27d5d, and the report says plainly which questions are still open: the human audit, the access decision, and the adjudication of the candidates.